Year 4 Science Revision: Chapters 1 to 4 [TEACHER'S KEY]

Student Name: ______________________ **Date:** ______________

Teacher: ANSWER KEY **Total Marks:** / 50

Section A: Multiple Choice Questions (15 Marks)

Circle or highlight the correct answer for each question.

1. What are the three main functions of the human skeleton?

A) Pumping blood, breathing, thinking
B) Support, protection, movement
C) Eating, sleeping, growing
D) Making food, storing water, running

2. Which of the following is an adaptation for an animal in a snowy habitat?

A) Large, thin ears to release heat
B) Webbed feet for swimming
C) Thick white fur and blubber
D) A long sticky tongue

3. In a food chain, a plant is always a:

A) Consumer B) Predator C) Prey **D) Producer**

4. When a muscle contracts to move a bone, what happens to it?

A) It gets longer and thinner
B) It gets shorter and thicker
C) It turns into a bone
D) It stays the exact same size

5. Which of the following statements about solids is true?

A) They can be easily squashed.
B) They take the shape of the container.
C) Particles are packed tightly in a pattern.
D) They can flow and be poured.

6. What is the process called when a liquid turns into a solid?

A) Melting **B) Freezing** C) Evaporation D) Condensation

7. Which of the following is a **natural** source of light?

A) A torch **B) The Sun** C) A mirror D) A glass window

8. How is a shadow formed?

A) When a transparent object lets light pass
B) When light bends around a corner
C) When an opaque object blocks light
D) When the sun goes down at night

9. Why do we get vaccines?

A) To cure a headache
B) To stop infectious diseases
C) To help bones grow faster
D) To digest food properly

10. Animals with a hard outer shell instead of bones have an:

A) Endoskeleton **B) Exoskeleton** C) Internal spine D) Organ shell

11. Which material allows almost all light to pass through it?

A) Wood B) Cardboard **C) Glass** D) Metal

12. Which of these is an example of an irreversible change?

A) Melting ice **B) Baking a cake** C) Freezing water D) Dissolving salt

13. What connects muscles to bones?

A) Joints B) Organs **C) Tendons** D) Skin

14. What do the arrows in a food chain represent?

A) Which animal is larger
B) The transfer of energy
C) The direction the wind blows
D) Where the animals live

15. What is the habitat of a camel?

A) Ocean B) Rainforest C) Arctic **D) Desert**

Section B: Fill in the Blanks (15 Marks)

Word Bank				
habitat	joints	liquid	friction	gas
energy	relax	fair	medicine	reflect
skull	opaque	transparent	bones	predator

Choose the correct word from the word bank above to complete each sentence.

1. The environment in which a plant or animal naturally lives is called its **habitat**.

2. All living things need **energy** to survive, which animals get from their food.

3. The bones in our head that protect our brain are called the **skull**.

4. Places where two bones meet, allowing us to bend and move, are called **joints**.

5. Muscles work in pairs; when one muscle contracts, the other must **relax**.

6. A material that does not let any light pass through it is called **opaque**.

7. A **liquid** is a state of matter that can flow and takes the shape of its container.

8. When conducting an experiment, keeping all conditions the same except for the one you are testing makes it a **fair** test.

9. Clear glass is an example of a **transparent** material because light passes clearly through it.

10. A force that slows down moving objects when they rub against each other is **friction**.

11. A drug prescribed by a doctor to help cure an illness is called **medicine**.

12. The human skeleton is made up of over 200 **bones**.

13. A state of matter where particles are spread very far apart and move freely is a **gas**.

14. Mirrors and shiny surfaces are able to **reflect** light back into our eyes.

15. An animal that hunts and eats other animals is known as a **predator**.

Section C: Structured Questions (20 Marks)

Write your answers clearly on the lines provided. There are 15 sub-questions in total.

Question 1: The Human Body & Medicines (4 Marks)
a) Name one major organ that the skull protects.

The brain.

b) Name two major organs that the ribcage protects.

The heart and the lungs.

c) Explain how the biceps and triceps work together to bend your arm at the elbow.

To bend the arm, the bicep contracts (gets shorter and thicker),

while the triceps relaxes.

d) What is an exoskeleton? Give one example of an animal that has one.

A hard outer shell that protects an animal. Example: Crab, spider, or insect.

Question 2: Ecosystems and Food Chains (4 Marks)
Look at the food chain below:

Grass → Grasshopper → Frog → Snake

a) Which organism is the **producer** in this food chain?

Grass.

b) Name one **predator** from the food chain above.

Frog OR Snake.

c) What does the arrow (→) in a food chain represent?

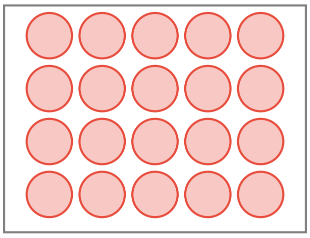
The flow or transfer of energy.

d) If a disease killed all the frogs in this habitat, what would likely happen to the population of grasshoppers? Explain your answer.

The grasshoppers would increase because

there are no frogs left to eat them.

Question 3: States of Matter (3 Marks)
a) In the box below, draw a simple model showing how particles are arranged in a **Solid**.



b) A student places a solid block of chocolate in the hot sun. What change of state will happen to the chocolate?

Melting (Changing from a Solid to a Liquid).

c) Is baking a cake an example of a reversible or irreversible change? Why?

Irreversible change, because a chemical reaction occurs

and you cannot get the raw ingredients back.

Question 4: Light and Shadows (4 Marks)
Sarah is using a torch to shine light on a wooden block. A shadow forms on the wall behind the block.

a) Why does the wooden block form a shadow? Use the word 'opaque' in your answer.

The wooden block is an opaque material, meaning it completely blocks

the light from passing through, creating a dark area (shadow) behind it.

b) Sarah wants to make the shadow on the wall **bigger**. What should she do with the wooden block?

Move the wooden block closer to the torch (or move the torch closer to the block).

c) If Sarah replaces the wooden block with a clear glass sheet, what will happen to the shadow?

There will be no shadow / a very faint shadow because glass is transparent and lets light through.

d) Does light travel in straight lines or wavy lines?

Straight lines.